

MSN ACADEMY

Virtual Internship Program 2.0

MSN TexWeave

BRD Audit Report

Team Identification

Field	Details
Project (Real Estate / Textile / Hospital)	Textile
Team Name	Team Nexus
Project Manager — Name & Intern ID	Sarim Nadeem
Team Member 1 — Name & Intern ID	Muhammad Bilal Shahid (INT-2026-075)
Team Member 2 — Name & Intern ID	Muhammad Ayaan (INT-2026-038)
Team Member 3 — Name & Intern ID	Alina Rania (INT-2026-068)
Team Member 4 — Name & Intern ID	Shafique Bhutto (INT-2026-059)

1. OBJECTIVE

The objective of the data profiling and cleaning phase was to assess the quality of the raw TexWeave dataset before loading into the database. The raw workbook sheets were analyzed and transformed using Power Query and Excel functions to identify and eliminate data quality issues, ensuring full consistency across all tables for database staging, SQL modeling, and Power BI dashboard reporting.

2. SUMMARY

The following data quality checks and cleaning transformations were performed:

- **Whitespace & Padding Removal:** Applied TRIM across text columns to remove leading, trailing, and irregular intermediate spaces.
- **Text Casing Standardization:** Standardized business codes/SKUs to uppercase (UPPER) and descriptive text/names to proper title case (PROPER).
- **Date & Timestamp Standardization:** Standardized all date attributes across every sheet to uniform ISO format (YYYY-MM-DD). For fields containing combined date and time strings (YYYY-MM-DD HH:MM:SS), timestamps were stripped out to retain clean, standardized date values.
- **Numeric & Currency Cleaning:** Removed special characters (currency symbols, commas) from price and cost fields and converted them to standardized numeric decimal formats.
- **Boolean & Default Flag Handling:** Standardized status and audit indicators (IsActive, IsObsolete) into binary integer formats (1 and 0) and populated missing default values.

3. DATA QUALITY AUDIT

#	Field(s)	Flaw Identified	Evidence / Count	How It Was Cleaned
Products_and_Finished_Inventory.xlsx — Products				
1.	ProductSKU	Leading/trailing whitespace and inconsistent lower/mixed casing	Multiple records with 'sku-001', 'PROD-004 ', 'prod-012'	Applied TRIM() and converted all SKU values to uppercase using UPPER().
2.	ProductName	Irregular spaces between words and inconsistent casing	Records such as ' slim fit jeans ', 'COTTON SHIRT'	Applied TRIM() to remove irregular padding and applied PROPER() for uniform Title Case.
3.	BasePrice, CostPrice	Formatted as text strings with currency signs and commas	Values containing "\$25.00", "PKR 1,500", and leading spaces	Removed currency symbols and commas, trimmed spaces, and changed data type to Decimal Number.
4.	UnitOfMeasure	Inconsistent naming conventions and abbreviations	Mixed entries: 'pcs', 'Pieces', 'PCS', 'piece'	Cleaned whitespace and standardized all variations to uniform 'Pcs'.

5.	CreatedAt	Combined date-time values and mixed date formatting	Records with timestamps like '2024-02-15 11:30:00' and serial numbers	Stripped out time components, extracted date only, and formatted to standard YYYY-MM-DD.
6.	IsActive	Blank and null cells	Empty cells in active product rows	Replaced blanks/nulls with default active flag value 1.
Products_and_Finished_Inventory.xlsx — BillOfMaterials				
7.	ProductSKU, MaterialCode	Trailing spaces and lowercase characters	Reference codes containing 'RAW-001 ' and 'prod-002'	Applied TRIM() and UPPER() across foreign reference columns to ensure clean relational joins.
8.	QuantityRequired	Stored as text values with trailing whitespace	Values stored as text strings (e.g., '2.50 ')	Trimmed spaces and converted column data type to Decimal Number (DECIMAL(18,4)).
9.	UnitOfMeasure	Inconsistent unit representations	Entries showing 'mtr', 'Meters', 'm', 'kg'	Standardized naming to uniform values ('Meters', 'KG') matching Raw Materials standards.
Products_and_Finished_Inventory.xlsx — DesignPatterns				
10.	PatternCode	Inconsistent casing and extra whitespace	Entries with ' pat-01 ', 'PAT-05'	Applied TRIM() to remove spaces and converted all pattern codes to uppercase via UPPER().
11.	PatternName, Colorway	Inconsistent text capitalization and padding	Lowercase names like 'dark indigo ', 'floral print'	Applied TRIM() and transformed values to Title Case using PROPER().
12.	IsObsolete, IsActive	Text boolean values instead of binary indicators	Values populated as 'TRUE', 'FALSE', 'Yes', 'No'	Replaced text booleans with standardized binary flags (1 for True/Yes, 0 for False/No).
Products_and_Finished_Inventory.xlsx — FinishedGoodsInventory				
13.	BatchNumber	Trailing spaces and mixed casing	Values containing ' batch-2024-01 '	Cleaned extra spaces using TRIM() and enforced uppercase

				standardization via UPPER().
14.	QuantityOnHand, ReservedQuantity	Stored with leading spaces and formatted as text	Numeric values stored as text characters	Trimmed whitespace and converted data type to whole numbers (Integer).
15.	LastUpdated	Timestamps and time values present with dates	DateTime strings like '2024-04-10 09:15:22'	Removed the time component, extracted the date part only, and formatted as YYYY-MM-DD.
Global Across All Workbooks & Sheets (Lookups, Purchasing, Manufacturing, Sales)				
16.	All Date Columns (OrderDate, DeliveryDate, StartDate, EndDate, DispatchDate)	Mixed date formats, serial numbers, and attached timestamps	Entries containing combined time values (HH:MM:SS) and slash- separated formats	Removed all timestamp/time portions completely across all sheets, extracted dates only, and standardized to YYYY-MM-DD.
17.	All Name & Category Fields (CustomerName, SupplierName, DepartmentName, StageName)	Hidden leading/trailing spaces and mixed casing	Unseen whitespace padding around entity names	Applied TRIM() across all descriptive text fields and standardized to Proper Case.
18.	All Code & Status Fields (CustomerCode, SupplierCode, PONumber, OrderNumber, Status)	Mixed lower/upper text casing and spacing	Inconsistent entries like 'pending', 'SO-001 '	Applied TRIM() and converted all business codes to uppercase (UPPER()) and status values to exact PascalCase.